SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: ITA 0603 **COURSE NAME: MACHINE LEARNING**

COURSE OUTCOME COVERED

CO2: *Neural Network Representation, Perceptron Model, Multilayer Perceptrons, Backpropagation Algorithm, Deep Neural Networks, Genetic Algorithms, Hypothesis Space Search, Genetic Programming, Models of Evaluation and Learning (BL1, BL2)*

Assessment Tool Weightage: 40%

QUESTIONS: 15 **Total Marks:25**

Annexure A

Analytical Problem Solving

Code of Conduct:

I Mohammed Mudhassir A/192424367 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Mudhassir

1. Perceptron Model (Student Pass/Fail)

Given:

- Weight $w_1 = 0.6$
- Weight $w_2 = 0.4$
- Bias $b = -0.5$
- Study Hours $x_1 = 2$
- Attendance $x_2 = 1$

Formula

$$Net = (w_1 x_1) + (w_2 x_2) + b$$

Calculation

$$\begin{aligned} &Net = (0.6 \times 2) + (0.4 \times 1) - 0.5 \\ &= 1.2 + 0.4 - 0.5 \\ &= 1.1 \end{aligned}$$

Step Activation Function

$$Output = \begin{cases} 1, & Net \geq 0 \\ 0, & Net < 0 \end{cases}$$

Since

$$1.1 > 0$$

Output = **1**

Conclusion

The perceptron predicts that the **student passes**.

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2. Backpropagation (Delta Rule)

Given

- Initial Weight = 0.5
- Learning Rate = 0.1
- Input = 2
- Target = 1
- Actual Output = 0.6

Step 1: Error

$$Error = Target - Output$$
$$= 1 - 0.6 = 0.4$$

Step 2: Weight Update

$$\Delta w = \eta \times Error \times Input$$
$$= 0.1 \times 0.4 \times 2$$
$$= 0.08$$

Step 3: New Weight

$$w_{new} = 0.5 + 0.08$$
$$= 0.58$$

Answer

- Error = **0.4**
- Weight Update = **0.08**
- New Weight = **0.58**

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3. Confusion Matrix

Given

- TP = 180
- TN =150
- FP =20
- FN =50

Accuracy

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$
$$= \frac{180 + 150}{400} = \frac{330}{400} = 0.825$$

= **82.5%**

Precision

$$Precision = \frac{TP}{TP + FP} = \frac{180}{200} = 90\%$$

Recall

$$Recall = \frac{TP}{TP + FN} = \frac{180}{230} = 78.26\%$$

Specificity

$$Specificity = \frac{TN}{TN + FP} = \frac{150}{170} = 88.24\%$$

Conclusion

- Accuracy = **82.5%**
- Precision = **90%**
- Recall = **78.26%**
- Specificity = **88.24%**

The model is fairly reliable, but the recall should be improved because false negatives are critical in medical diagnosis.

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4. Genetic Algorithm

Given

Population:

$$\{2, 4, 6, 8\}$$

Fitness Function

$$f(x) = x^2$$

Fitness

Individual	Fitness
2	4
4	16
6	36
8	64

Best candidates:

- 8
- 6

Binary Representation

8 = **1000**

6 = **0110**

Crossover (Swap LSB)

Before

1000

0110

Swap last bit

1000

0110

(Last bits are both 0, so offspring remain unchanged.)

Offspring:

- 1000 = 8
- 0110 = 6

Mutation

Mutation randomly changes one or more bits to:

- increase diversity
- avoid local optimum
- prevent premature convergence

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5. Training vs Validation Accuracy

Observation

Training Accuracy

60% → 95%

Validation Accuracy

58% →72% →68%

Issue

Overfitting

Reason

The model memorizes training data instead of learning general patterns.

Solutions

1. Early Stopping
2. Dropout/Regularization

Importance of Validation Accuracy

Validation accuracy measures how well the model performs on unseen data and helps detect overfitting.

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6. Perceptron (Loan Approval)

Given

- $w_1 = 0.7$
- $w_2 = 0.5$
- $b = -0.6$
- Income=1.5
- Credit Score=2

Net Input

$$\begin{aligned} &= (0.7 \times 1.5) + (0.5 \times 2) - 0.6 \\ &= 1.05 + 1 - 0.6 \\ &= 1.45 \end{aligned}$$

Since Net>0

Output=1

Conclusion

Loan Approved

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7. Delta Rule

Given

Weight=0.4

Learning Rate=0.2

Input=3

Target=0.9

Output=0.5

Error

$$0.9 - 0.5 = 0.4$$

Weight Update

$$0.2 \times 0.4 \times 3 = 0.24$$

New Weight

$$0.4 + 0.24 = 0.64$$

Answer

- Error = **0.4**
- Weight Update = **0.24**
- New Weight = **0.64**

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8. Confusion Matrix

Given

TP=120

TN=200

FP=30

FN=50

Accuracy

$$= \frac{320}{400} = 80\%$$

Precision

$$= \frac{120}{150} = 80\%$$

Recall

$$= \frac{120}{170} = 70.59\%$$

Specificity

$$= \frac{200}{230} = 86.96\%$$

F1 Score

$$= \frac{2PR}{P + R}$$
$$= \frac{2(0.8)(0.7059)}{0.8 + 0.7059} = 75\%$$

Conclusion

The recall is only **70.59%**, making the model less suitable for applications where false negatives are critical.

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9. Genetic Algorithm

Population

{1,3,5,7}

Fitness

$$f(x) = 2x + 3$$

x	Fitness
1	5
3	9
5	13
7	17

Best:

- 7
- 5

Binary

7=111

5=101

Single-point crossover

111

101

↓

101

111

Offspring

5 and 7

Mutation

Mutation introduces random changes that help avoid premature convergence and maintain diversity.

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10. Training Loss vs Validation Loss

Observation

Training loss decreases continuously.

Validation loss decreases then increases.

Issue

Overfitting

Reason

The network memorizes the training data.

Solutions

- Early Stopping
- Regularization (L1/L2)

These improve generalization on unseen data.

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11. Perceptron (Customer Purchase)

Given

Weights

0.5

0.3

Bias

−0.4

Inputs

2

1

Net Input

$$
\begin{aligned}
&= (0.5 \times 2) + (0.3 \times 1) - 0.4 \\
&= 1 + 0.3 - 0.4 = 0.9
\end{aligned}
$$

Output=1

Conclusion

Customer Purchases the Product

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12. Delta Rule

Given

Weight=0.6

η=0.15

Input=2.5

Target=1

Output=0.7

Error

$$1 - 0.7 = 0.3$$

Weight Update

$$0.15 \times 0.3 \times 2.5 = 0.1125$$

New Weight

$$0.6 + 0.1125 = 0.7125$$

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13. Confusion Matrix

TP=140

TN=180

FP=25

FN=35

Accuracy

$$= \frac{320}{380} = 84.21\%$$

Precision

$$= \frac{140}{165} = 84.85\%$$

Recall

$$= \frac{140}{175} = 80\%$$

Specificity

$$= \frac{180}{205} = 87.80\%$$

F1 Score

$$= \frac{2(0.8485)(0.8)}{1.6485} = 82.4\%$$

Conclusion

The model is suitable for fraud detection due to its high precision, reducing costly false positives.

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14. Genetic Algorithm

Population

{2,5,7,9}

Fitness

$$f(x) = 3x + 2$$

x	Fitness
2	8
5	17
7	23
9	29

Best

9 and 7

Binary

9=1001

7=0111

Single-point crossover

1001

0111

↓

1001

0111

(Offspring remain unchanged if crossover point does not alter differing segments; another crossover point could generate different offspring.)

Mutation

Mutation increases population diversity, prevents stagnation in local optima, and improves exploration of the search space.

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15. Training vs Validation Accuracy

Observation

Training Accuracy

65% →97%

Validation Accuracy

63% →75% →72%

Issue

Overfitting

Reason

The model learns noise and specific patterns from the training data, reducing performance on unseen data.

Solutions

- Dropout
- Data Augmentation